

Manual

Shopfloor Monitor (Graphic Machinery) MDE-SFM 8.1

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1 Overview of Graphic Machinery

Purpose

The graphic machine park provides a visual presentation of the company's machine park at a specified level (workshop, foreman's area, ...). Not only does this function visualize master data, but it also presents in real time actual data recorded in MDE and BDE.

Implementation considerations

You use the function package if:

- You would like to have a graphical presentation of your company's available machines that is supplemented by current actual data.

Integration

Current MDE data (e.g. machine status, current quantities) and BDE data (e.g. logged on operations) are accessed for the presentation. Additionally, there is an option to display the current status of the terminal or ZKS accesses.

Features

- Graphical presentation
 - Graphical presentation of the company's machine park in a hall, department, foreman's area or the like.
- Two-level structure
 - Two-level structure from separate machines and their assignments to machine groups, cost centers and lines.
- Display of current machine state
 - Display of the current machine state using different colors
- Tool tip
 - Detailed information in the info window (Tool tip)
- Quantity progress
 - Bar chart provided to visualize the finished actual numbers of units compared to the planned target number of units.
- Layouteditor
 - Graphic editor used to create hall layouts with individual placement of the machine symbols, machine groups and production lines.

2 Graphic Machinery

Overview

Menu	Production facility management → Current information → Graphic machinery Information management → Production overview → Graphic machinery Human resources management → Access control → Security control center
Transaction code	mpark
Function authorization	mpark

Usage

The graphic machinery provides a vivid presentation of essential workplace and production information in spatial/temporal contexts. The user can create and display different representations (so-called layouts) of all organizational units (e.g. halls, departments) as required.

Graphic machinery can serve as an indicator for the production manager and also provides a current, large-screen overview of production statuses on the shop floor.

Integration

In the graphic machinery, you can visualize up-to-date status information of objects from various HYDRA products:

- BDE/MDE: Machines, groups, line groups
- MW: Terminals
- WRM/EMG: Resources (no file-based resources, no DNC resources)
- MPL: Material buffers
- ZKS: Accesses, access groups

Requirements

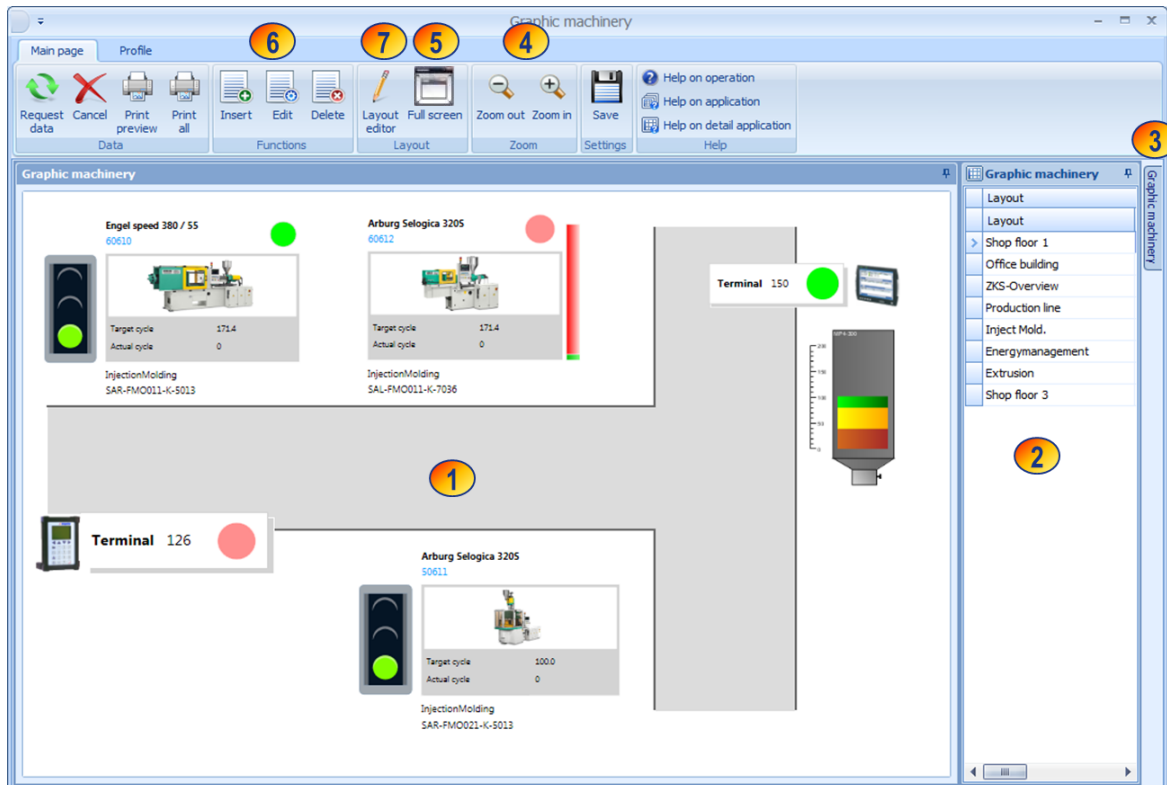
The relevant objects to be visualized in the graphic machinery must be set up in the system master data.

Appropriate layouts must have been prepared using the [layout editor](#).

Structure of graphic machinery

The graphic machinery shows various layouts of different organizational units. These may be generated by the user as required.

To navigate through the layouts, you can use the table on the right hand side (2). The table contains the layout names and additional information relating to the individual layouts. By left clicking a table row, you can switch to the relevant layout which is then displayed in the layout area (1).



- (1) Layout area
- (2) List of layouts created
- (3) Detail panel with information on the currently selected layout.
- (4) Zooming in and out displayed layouts
- (5) Activation of full screen mode; canceled by "Esc" button.
- (6) [Editing functions](#) for creating a new layout and editing or deleting an existing layout.
- (7) Function to [call up](#) the [layout editor](#) for graphic editing of the actual layout.

The statuses of machines, groups, terminals, etc. are updated at regular intervals (every 30 seconds).



You can only request the Help function via F1 when the list of created layouts is selected. In other cases, please call up the Help function using the [Help on application](#) button.

Operation

The table on the right side of the screen (2) is used for navigating through the graphic machinery. By clicking a table row, you switch to the requested layout.



The width of the right table can be saved user-specifically by clicking the "Save settings" button. It is not possible to hide the table completely. Tip: The table can be "folded away" to the side using the respective button. You can find additional information on general operation in the documentation "[Operation of MES Operation Center](#)".

If you right click an icon in the layout area (1), a context menu opens where you can switch to other applications, provided that you have the required authorization.

When hovering with the mouse pointer over a template in a specific layout, a tooltip specific to each object opens showing additional information. The display time cannot be configured.

Field Descriptions

The following field descriptions refer to the "Layout" table and the "Layout" detail view:

Layout

Layout name, e.g. "Hall 1"

Comment

Comment further describing the layout.

Order

By determining an order/sequence and sorting the table by the "order" column, a particular layout is always displayed by default when the function is called up. Please note that the defined order is not user-related.

Responsibility area

A responsibility area can optionally be assigned to a layout. In this case, the layout will only be available to a user with authorization for the entered responsibility area.

Internal ID

Upon insertion of any new layout, the system assigns this number automatically; it cannot be modified by the user.

File name

The configuration of the actual layout is saved in a file in the server. The system automatically assigns the file name when a new layout is added and cannot be modified by the user.

Type

Always "N"

Modified by / Modified on

This information refers to the latest editing of a layout entry in the table, not to the latest editing of the graphic layout.

Toolbar



Insert

Function authorization: mparkle.create

To create a new layout, you have to insert a new entry first. Please provide the following information:

- Layout: Layout name
- Comment: Description
- Order: the list can be sorted by this column
- Responsibility area (optional)



Edit

Function authorization: mparkle.edit

You may change the following data in an existing layout

- Layout
- Comment
- Order
- Responsibility area.



Delete

Function authorization: mparkle.delete

Upon confirmation, the layout entry is deleted from the database.

Please note: The file storing actual layout information in the server is not deleted.



Layout editor

Function authorization: mparkle.layedit

Calls up the [layout editor](#) for the entry currently selected in the list.



Full screen

Zooms in the active layout to the full screen size. Press "Esc" to cancel the full screen view.



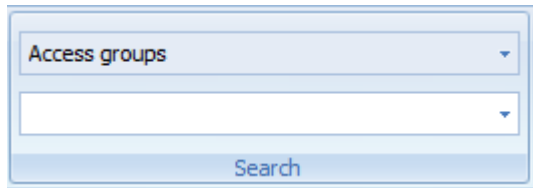
Zoom out

Zooms out the current layout in the application.



Zoom in

Zooms in the current layout in the application.

**Search**

These two drop-down list boxes can be used to search for specific objects in layouts. The object type (workplaces/machines, groups, line groups, material buffers, resources, terminals, accesses and access groups) can be selected in the upper field. The number of the searched object can be selected in the lower field.

Once the object has been selected, the application highlights all layouts containing the relevant object in the list of available layouts and shows one of these layouts automatically. The searched object is highlighted with a red frame.

When a layout is saved, the application identifies the objects included in the layout. Older layouts might have to be saved once more in order to select and highlight the included objects.



The function "searching for objects" is only available if the modification [GraphicMachinery2Search](#) is enabled.



The graphic machinery does not support the functions  "Print preview" and  "Print all".

Selection Criteria

Selection criteria are not available in the graphic machinery. When the application is called up, all configured layouts are loaded automatically and the first layout in the list is displayed.



By defining the layout order and sorting the list by the "order" column, a particular layout is always displayed by default when the function is called up. Please note that the defined order is not user-related.

Additional features of the graphic machinery

Opening full screen layouts

The menu editor allows creating an entry opening a defined layout in normal view or full-screen mode in the menu of a specific MOC client (menu depending on the user --> USER scope).



This function is only available if the menu editor and the corresponding function authorization "mparklink" are used.

Please proceed as follows if the menu editor is used:

1. Open the menu editor:

Menu --> Extras --> Menu editor

2. Drag the graphic machinery to the required menu and change values (enable advanced settings)

- **Caption:** can be selected individually, e.g. Mpark layout 1
- **ID:** GraphicMachinery2
- **Command:** mpark
- **Parameter:** fullscreen id=Internal ID of the layout
The parameter "fullscreen" opens the defined layout in full screen mode.

Example: "Layout 1" has the internal ID 87 (see screenshot)

The top screenshot shows a table with the following data:

Layout	Commentary
energy management	Energymanagement - Shop floor 2
Extrusion	Extrusion
InjectionMolding	Inject Mold. Shop Floor 1
Office building	Office building Floor 2
Production line	Shop floor 1
Shop floor 1	Production hall 1
Shop floor 3	Shop floor 3
ZKS-Overview	Shop floor 4 - Accesspoints

The bottom screenshot shows the configuration fields for a selected layout:

Internal ID	27
Layout	Shop floor 1
Commentary	Production hall 1
Order	1
Responsibility area	

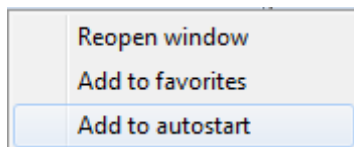
Menu Editor

Command	Labeling	ID	Command	Parameter
MparkLayout2	MparkLayout2	GraphicMachinery2	mpark	fullscreen id=27



If the "autostart" function is used, you can open a defined layout in full screen mode when starting the MOC.

Autostart: right click and choose "add to autostart"



Setting the refresh interval

The refresh interval refers to the time interval between two update runs in the graphic machinery. This interval can be set customer-specifically via an INI data configuration.

The value corresponds to the time interval in milliseconds between two update runs.



The default value of 60000 milliseconds (60 seconds) applies, if a value less than 60000 milliseconds (60 seconds) is entered or if nothing is entered.

Configuration:

Ini configuration name:	MPARK
Section:	MPARK2
Key:	REFRESHRATE
Value:	240000 (corresponds to 4 min)

Disabling offline messages

If the graphic machinery cannot establish a connection to the server (offline), the application displays an error message every time you request data.

You might not want this behavior, especially if you use the full screen mode. Configure the [INI configuration](#) as follows to prevent the error message from being displayed:

Configuration:

INI configuration name:	MPARK
Section:	MPARK2
Key:	SUPPRESSCONNECTIONERRORMESSAGE
Value:	y
	Default value: n

If you start the graphic machinery the next time and you cannot request data (as a connection to the server cannot be established), the system only records entries in the MOC [Log window](#).

But to inform the user, the window title of the full screen indicates the note "(Offline)" behind the layout name.

3 Graphic Machinery - Layout Editor

Overview

Menu	Production facility management → Current information → Graphic machinery Human resources management → Access control → Security control center Information management → Production overview → Graphic machinery
Transaction code	mpark
Function authorization	mparkle.layedit
Function key	 Layout editor

Usage

The layout editor of the [graphic machinery](#) enables the user to "design" so-called layouts for freely definable organizational units (e.g. halls, sections, other units, etc.).

Integration

The layout editor is integrated in the [graphic machinery](#) application and is called up from there using the

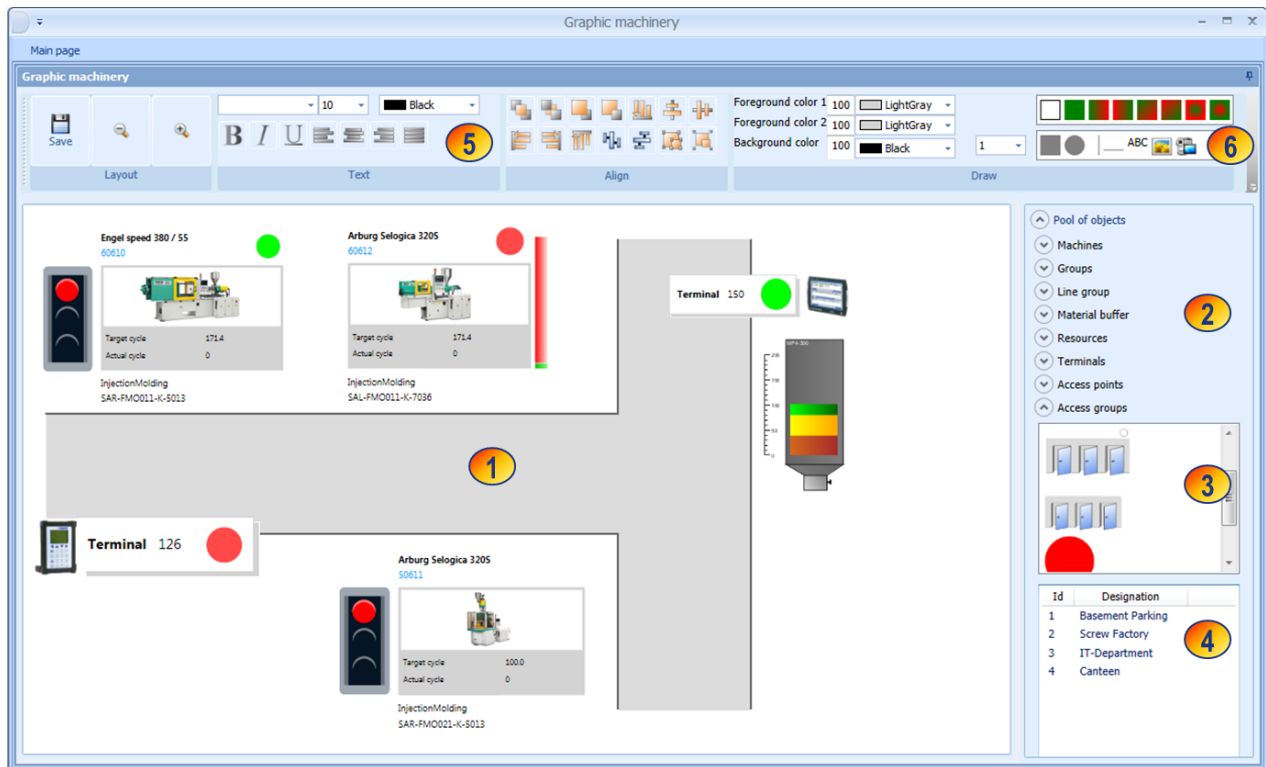


function key.

Prerequisite

The relevant objects to be displayed in a layout have to be set up in the system master data. A layout entry must have been defined in the [graphic machinery](#) .

Layout editor



- (1) Layout area
- (2) Pool of objects: by clicking on an object type (machine, ...), the following is displayed below:
- (3) Templates of the respective object type
- (4) Available objects (machines, ...)
- (5) In the upper area, the "tools" for graphic editing of a layout are shown (the illustration shows an example; the precise icons available are described in the [toolbar](#) section).
- (6) Graphic objects - in addition to the templates - are square, circle, lines, text, images, video and/or video stream (mms).



It is not possible to activate the Help function via F1 in the layout editor. Please activate the Help function using the *Help on application* button.

Toolbar



Please note that there is no Quick launch bar in the layout editor and the toolbar cannot be minimized, either.

Furthermore, the layout editor does not support any functions of the MES Development Suite.

The two functions  "Print preview" and  "Print all" are not supported in the layout editor.

Layout category



Save

Press "Save" if you have made changes in the layout.



A confirmation prompt will appear when you exit the application if you have previously made changes.



Zoom out layout in layout editor.

Please note: a change in size in the layout editor will not affect the display in the graphic machinery.



Zoom in layout in layout editor.

Please note: a change in size in the layout editor will not affect the display in the graphic machinery.

Text category

The functions described below can be used to format a static text which was transferred to the layout via

drag & drop using the button .



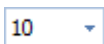
Changes always refer to the entire text within a text field:

Please note that

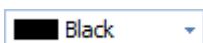
- the buttons do not show the current format of a text and
- changing the format (e.g. to *italics*) will also reset the font, font size and color at the same time.



Changes the font for a static text.



Changes the font size for a static text.



Changes the font color for a static text. The entire text in the text field is changed.



Changes the text to bold.

In order to display the text normally again, click on the button again.



Changes the text to italics.

In order to display the text normally again, click on the button again.



Underlines the text.

In order to display the text normally again, click on the button again.



Aligns a static text, left-justified:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwiewoi



Aligns a static text, centered:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwiewoi



Aligns a static text, right-justified:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwiewoi



Aligns a static text, fully-justified:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwiewoi

Alignment category

You may use the functions in this category to align graphic objects.



Brings the currently selected element to the foreground.



Brings the currently selected element to the background.



Brings the currently selected element one level further to the front.



Brings the currently selected element one level further to the back.



Aligns the selected elements to the left.



Aligns the selected elements to the right.



Aligns the selected elements to the top.



Aligns the selected elements to the bottom.

Drawing category

The functions described below can be used to format a graphic object which was transferred to the layout via drag & drop using the  button.



Changes always refer to all selected objects.

Please note that

- the buttons do not show the current format of a graphic object and
- changing the format (e.g. change of color gradient direction) will also reset the color.

Background color



Changes the color of the objects without color gradient.

Changes the first color of objects with color gradient.

For text fields, this is the background color; the foreground color selection is made in the text category.

2nd Background color  LightGray ▾



Changes the second color of objects with color gradient.
For text fields, this is the background color; the foreground color selection is made in the text category.

Line/frame color  Black ▾



Changes the line color.



Changes the color of the frame around the object.

1 ▾



Changes the thickness of a line.



Determines the filling effect - filling or color gradient.
For text, this is the background color.

First select a filling effect by clicking with the left mouse button before drawing one of the objects 



Click on this icon if you intend to draw a quadrangle (rectangle, square). You transfer this into the layout by drag & drop.



Click on this icon if you intend to draw a circle or an oval. You transfer this into the layout by drag & drop.



Click on this icon if you intend to draw a vertical line. This is transferred into the layout by drag & drop in the defined thickness (see above).

When dropping in the layout area, a square is displayed which can then be modified in size and shape as long as the object is selected.

Please note: drawing diagonal lines is not possible.



Click on this icon if you intend to draw a horizontal line. This is transferred into the layout by drag & drop in the defined thickness (see above).

When dropping in the layout area, a circle is displayed which can then be modified in size and shape as long as the object is selected.

Please note: drawing diagonal lines is not possible.



Click on this icon if you intend to enter (static) text. This is transferred into the layout by drag & drop in the defined font type, font size, color and/or background color (see above).

When dropping in the layout area, a text field with the letters "ABC" is displayed. This text can then be modified as long as the object is selected.



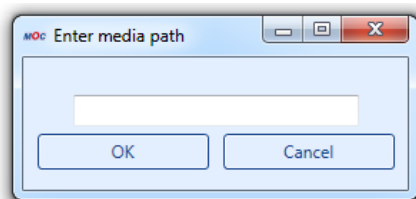
This icon can be used to insert an image into the layout. When dropping in the layout area, an "Open file" dialog is displayed from which you can select and transfer the desired image.

The following formats are supported: jpg, png, bmp, gif.

Please note: The image is transferred to the layout, i.e. into the zip file of the layout. The file size will therefore have an effect on the loading time of the layout. It is recommended to use images in compressed file formats, whose file size are in a double-digit and/or maximum three-digit kilobyte range.



This icon can be used to integrate videos or video streams into the layout. Select the icon, drag it into the layout and drop it there. A dialog opens with an input field in which you enter the file path of the video and/or video stream.



The following formats are supported:

- Video: mp4, wmv

The file path may be defined absolutely or relatively in relation to the MOC directory.

- Video stream: mms

The MMS protocol (Microsoft Media Server Protocol) is a protocol developed by Microsoft and serves the transmission of multimedia streams. The stream has to start with mms://, e.g. mms://192.168.50.10:8080/

Please note: It is not the video which is transferred into the layout, but only the path to the video. For this reason, please make sure that the video has been saved at a central location accessible from all MOC Clients.

Operation

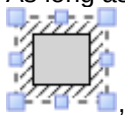


Please note that there is no undo function in the layout editor.

Transfer of a graphic object - square, line, text, image, ... - into the layout

The "placing" principle for a graphic object in the layout area is as follows:

1. If you wish to add a square, a circle or text, first select the filling effect by clicking with the left mouse button: . The selected icon is highlighted in color.
2. Select an object by clicking on the left mouse button: . The selected icon is highlighted in color.
3. Now drag this object with the left mouse button pressed down into the layout area and drop it at the desired location (drag & drop).
4. If the object is an image or video/video stream, a dialog for selecting the image and/or an input field for entering the file path for the video/the video stream will appear.
5. As long as the object is selected



you can modify its properties and/or transfer it to another location by drag & drop.

Selecting an object in the layout

In order to select an individual graphic object, click on it. A "frame" will appear around the object. An example with a square is shown below:

Before (before clicking):



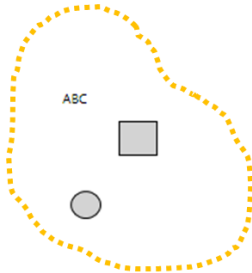
After (after clicking):



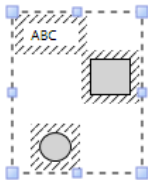
Selecting several objects in the layout

In order to select several objects at a time, "move" around the desired objects with the left mouse button pressed down, so that an orange, dotted circle is formed. Release the left mouse button when you have finished:

Before releasing the left mouse button:



After releasing the left mouse button:



Format changes, e.g. a change in (background) color, will now have an effect on all objects selected.

Transfer of an object (machine, group, ...) into the layout

In order to visualize an object, e.g. a machine, a group or an access in the layout, proceed as follows:

1. Make sure that the object types are displayed in the detail panel on the right. If in doubt, click on the text "Pool of Objects" so that all object types are shown below (the number is displayed in the [overview graphic](#) at the top; it is intended for orientation only and is not available in the application):

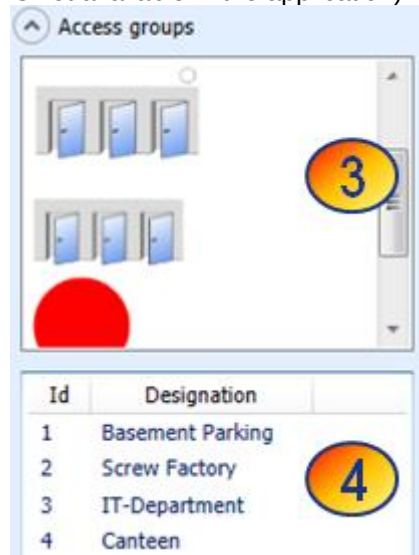


2. Now click on the object type for which you intend to transfer an object into the layout. Two windows are shown directly beneath the object type designation:

top (3): the templates available for this object type

bottom (4): the objects of this object type available in HYDRA

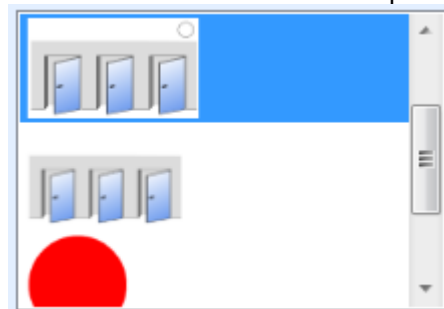
(the numbers are displayed in the [overview graphic](#) at the top; this is intended for orientation only and is not available in the application):



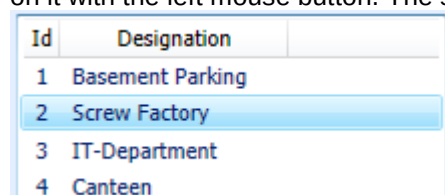
Wait a few seconds if only IDs but no designations are displayed in the selection window at the bottom.

The number of columns as well as the column titles in the selection window at the bottom depend on the object type.

3. Now select a template which is to be displayed for the object in the layout by clicking on it with the left mouse button. The selected template is highlighted in color.



4. In the bottom selection window, now select the object which is to be visualized in the layout by clicking on it with the left mouse button. The selected object is highlighted in color.



5. Now drag this object with the left mouse button pressed down from the selection window at the bottom into the layout area and drop it at the desired location (drag & drop).
6. Click on the object in the layout if you wish to change its size or drag & drop it to another location.



Templates in graphic machinery

The graphic machinery provides appropriate templates for the object types listed below.

- Workplaces/machines
- Groups (capacity groups)
- Lines (workplaces/machines with type "L")
- Material buffers
- Resources (no DNC and file resources)
- Terminals
- Accesses
- Access groups

Existing templates can be modified in the course of customizing.

Layout deployment

Completed layouts can be transferred from one system to another. The transfer procedure is demonstrated in a corresponding service.